BALAKUMARAN M

PCB & Hardware Design Engineer | Fresher | Immediate Joiner

+91-7010125244 | balakumaran2580@gmail.com | Salem, India | GitHub: github.com/Bala-Kumaran

SUMMARY

Enthusiastic and motivated B.E. Electronics & Communication Engineering graduate with hands-on experience in PCB Design, PCB Layout, Schematic Design, Hardware Testing, and Hardware Validation using KiCad. Skilled in component selection, PCB routing, ERC/DRC, design verification, Gerber generation, and technical documentation. Seeking an entry-level Trainee – PCB Design Engineer opportunity and immediate joiner.

EDUCATION

B.E. Electronics & Communication Engineering | Dhirajlal Gandhi College of Technology | Salem | 2022–2026
CGPA: 7.53/10 | Coursework: Electronic Circuits, Analog Electronics, Digital Electronics, Microprocessors, Power Electronics, Circuit Analysis

INTERNSHIP

PCB Testing & Design Validation Intern | Akira Controls, Coimbatore | May 2026 – July 2026
• Performed board-level functional and continuity testing on 15–20 servo motor controller PCBs daily, supporting hardware validation and defect identification.
• Executed design verification using ERC checks, circuit troubleshooting and solder-joint inspection to identify PCB and assembly-related defects.
• Collaborated with senior engineers on fault isolation and circuit debugging to support hardware reliability and improve test coverage.
• Prepared technical documentation and test reports for hardware validation and quality-assurance activities.

PROJECTS

Mini Inverter PCB Design | KiCad
• Created schematic and PCB layout with component selection, footprint assignment, component placement and PCB routing.
• Performed ERC/DRC verification and generated Gerber files for PCB fabrication.

5V Regulated Power Supply PCB Design | KiCad
• Designed schematic and PCB layout with component selection, placement, routing and design verification.
• Created BOM, Gerber files and PCB fabrication documentation.

Smart Battery Management System for EV
• Developed an embedded monitoring system for real-time voltage, current, and temperature tracking using digital sensors and microcontrollers.
• Implemented digital threshold logic to trigger automated safety cut-offs, protection circuits, and IoT alerts

Drowsiness Detection System
• Designed sequential logic state flows to process real-time sensor inputs for immediate operator drowsiness detection.
• Integrated digital output logic to control synchronized hardware alerts including LEDs, buzzers, and communication modules.

TECHNICAL SKILLS

PCB Design & EDA: KiCad, EasyEDA, Schematic Capture, PCB Layout, Component Selection, Footprint Assignment, Component Placement, PCB Routing
Design Verification: Electrical rule checking (ERC), design rule checking (DRC), schematic-to-layout verification, circuit analysis, design verification methodologies.
PCB Fabrication & Documentation: Gerber file generation, Bill of materials (BOM) creation, fabrication documentation, technical specifications, design rules, manufacturing considerations
Hardware Testing & Validation: Board-Level Testing, Functional Testing, Continuity Testing, Hardware Validation, Circuit Debugging, Fault Detection, Defect Analysis, Prototype Testing, Failure Analysis
Electronics: Analog Electronics, Digital Electronics, Power Electronics, Power Supply Design, Electronic Components, Component Datasheets

CERTIFICATIONS & ACHIEVEMENTS

Seminar: Advanced PCB Design using KiCad: Schematic to Fabrication — Vellore Institute of Technology, Chennai | 2025
Certificate of Appreciation – BIS Activities & Community Engagement | Dhirajlal Gandhi College of Technology | 2025